

Paper Code: ITC-101
Examination –Dec- 2024

B.Tech. I Sem :

Subject name: Python Programming

Max. Marks : 60

Min. Marks : 19

total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

word limit be observed as follows:

Part (a)(i) and a(ii) Objective type/Fill in the blanks/true& false.

Part (b) – Three/four sentences based answer question (Maximum 50 words)

Part (c) – Short Descriptive Ans./Numerical/ algorithm /programs based Question.
 (Maximum 100 words)

Part (d) – Long Answer /Shorts -notes /Descriptive Ans. based Question (Maximum 400 words)
 word limit NOT to be followed for diagram, numerical, derivation.

	Marks	BL	CO
a)-I <pre>>>>a = 10 >>>b = a >>>b += a >>>print(b)</pre> The output for the above lines of code is _____.	01	1	1
(a)-II To write multi-line comments you can use '#' at each line or you can use _____	01	1	3
(b) Write down the features of Python.	02	2	2
(c) What is the difference between mutable and immutable data types? Explain with example.	03	3	2

OR

What do you mean by type casting? Discuss the explicit type casting with example.	03	3	2
(d) With the help of examples, explain the various data types used in Python.	05	5	2

OR

Discuss the following:

(i) Indentation

(ii) Escape Sequence Characters

05	5	3
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Q.2 (a)-I The continue statement _____ the rest of the loop statements and causes the next iteration to occur.	01	1	2
(a)-II The conditional operator != is _____.	01	1	2
(b) Explain and write the output for the following: <pre>for k in range(-6,-6,2) print(k)</pre>	02	2	2

- (c) With the help of an example, explain the python match case statements. 03

OR

Discuss the following:

- (i) Break statement
- (ii) Nested loops

- (d) Write a program to test whether a number is divisible by 5 and 10 or by 5 or 10. 05

OR

Using for loop, write a program to create the multiplication table of 5. 05

Q.3 (a)-I >>>str = "Silver Spoon"
>>>print(str.split(" "))

The output for the above lines of code is _____.

(a)-II The seek () function allows to move the _____ within a file to a specific position. 01

(b) What do you mean by strings? How can you loop through a string? 01

(c) Write the output for the following: 02

pie = "ApplePie"

print (pie[:4])

print (pie[-7:])

print (pie[2:5]) 03

OR

How are numbers written in a file? Explain with example. 03

- (d) Given a CSV file named data.csv, write a Python program to read the contents of the file and display them. 05

OR

Explain the following:

- (i) os Module
- (ii) File handling

Q.4 (a)-I list.sort() method sorts the list in _____ order. 01

(a)-II We can check if a given item is present in the tuple. This is done using the _____ keyword. 01

(b) What is a list? Explain its properties. 02

(c) Explain with an example how a tuple is different from a list.

03

3	2
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OR

What do you mean by traversing dictionaries? Explain.

03

5	2
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(d) Write the output for the following expressions.

05

- ```
(i) >>> L1=['A','B',3,4,5]
>>> L1[: :-1]
>>> print(L1)
(ii) >>> L1.pop(4)
>>> print(L1)
(iii) >>> Grades = {"Sahil":90,"Abhijeet":65}
>>> print(len(Grades))
(iv) >>> del Grades["Abhijeet"]
>>> print(Grades)
(v) >>> t = (1, 2, 3, 4)
>>> t[2] = 10
>>> print(t)
```

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|---|---|
| 5 | 2 |
|---|---|

OR

Given a dictionary `fruit_prices = {'apple': 50, 'banana': 30, 'cherry': 60}`, write a program to:

05

- Add a new key-value pair for 'orange' with a price of 40
- Remove the key-value pair for 'banana'
- Update the price of 'apple' to 55
- Print all keys and values in the dictionary

|   |   |
|---|---|
| 5 | 2 |
|---|---|

2.5 (a)-I A \_\_\_\_\_ is a blueprint or template for creating objects.

01

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|---|---|
| 1 | 3 |
|---|---|

(a)-II In python we can raise custom errors by using \_\_\_\_\_ keyword.

01

|   |   |
|---|---|
| 2 | 3 |
|---|---|

(b) Define a class in Python. Explain with example.

02

|   |   |
|---|---|
| 2 | 3 |
|---|---|

(c) Differentiate between method overloading and method overriding in Python.

03

|   |   |
|---|---|
| 3 | 3 |
|---|---|

OR

Explain the following:

03

|   |   |
|---|---|
| 3 | 3 |
|---|---|

- (i) Constructors
- (ii) self parameter

(d) What is an exception in Python? Write a program to demonstrate exception handling using the try-except-finally structure.

05

|   |   |
|---|---|
| 3 | 3 |
|---|---|

OR

Explain the concept of inheritance. Explain how a child class can inherit properties and methods from its parent class.

05

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|---|---|
| 5 | 3 |
|---|---|

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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